

PROBLEM STATEMENT

Recommend safe and efficient routes based on real-time and historical conditions.

Industry Context

This concept fits within the **Intelligent Transportation Systems (ITS), smart mobility, and navigation technology industry**. Modern transportation platforms use real-time data such as traffic congestion, road incidents, weather, road closures, and vehicle speeds, combined with historical traffic patterns, to recommend routes that balance **travel time, safety, reliability, and fuel efficiency**.

It is particularly relevant to:

- **Navigation & mapping:** Dynamic route recommendations for commuters and travelers.
- **Logistics & fleet management:** Optimizing delivery and commercial vehicle routes.
- **Ride-hailing & mobility:** Helping drivers and passengers choose efficient routes.
- **Public transportation:** Improving routing and reducing delays.
- **Emergency response:** Identifying safer and faster routes for ambulances, fire services, and police.
- **Smart cities:** Using connected-road and traffic data to improve overall urban mobility.
- **Autonomous and connected vehicles:** Supporting real-time decision-making using road, traffic, and environmental data.

Problem Statement

Navigation systems often recommend routes based mainly on distance and estimated travel time. However, real-world conditions such as traffic congestion, accidents, road closures, weather, and unsafe road segments can change quickly. There is a need for a system that combines real-time and historical data to recommend routes that are both safe and efficient.

Existing route-planning systems may not adequately integrate **real-time traffic conditions with historical travel and incident data**. This can result in route recommendations that become inefficient or unsafe when road conditions change. The proposed system aims to develop a dynamic route recommendation approach that analyzes multiple data sources to identify routes based on **travel time, congestion, incidents, road conditions, and historical patterns**.

Rapid urbanization has increased traffic congestion, travel delays, and road-safety challenges in cities. Conventional navigation systems primarily focus on reaching destinations quickly, without sufficiently considering safety and changing environmental conditions. A smart routing solution is required that uses **real-time traffic, historical patterns, weather, and road-risk information** to support safer and more efficient urban transportation.

Objectives

- To develop an intelligent route recommendation system that considers both real-time and historical transportation data.
- To analyze real-time factors such as traffic congestion, accidents, road closures, weather, and road conditions.
- To identify historical traffic patterns and use them to predict potential delays and congestion.
- To recommend routes that provide a balance between safety, travel time, and efficiency.
- To dynamically update route recommendations when road conditions change.
- To reduce travel delays, fuel consumption, and unnecessary detours.
- To improve the overall reliability and safety of navigation and transportation.
- To provide users with data-driven route recommendations for better travel decision-making.

Expected Outcomes:

1. Develop an end-to-end Intelligent Route Recommendation System for collecting, processing, and analyzing real-time and historical transportation data.
2. Provide a GUI/dashboard for viewing routes, traffic conditions, road incidents, and recommended travel paths.
3. Maintain information such as traffic density, travel time, road conditions, accident records, weather conditions, road closures, historical traffic patterns, and route risk indicators.
4. Enable contextualization and correlation of real-time and historical data to identify traffic patterns, potential delays, and road-safety risks.
5. Provide dynamic and efficient route recommendations that consider travel time, congestion, safety, reliability, and changing road conditions.
6. Allow users to view and compare multiple route options along with relevant information such as estimated travel time, distance, traffic conditions, and potential risks.
7. Apply key software engineering principles such as security, scalability, maintainability, reliability, usability, and performance.
8. Demonstrate proper requirements analysis, system design, SDLC selection, data processing, route optimization, and evaluation of system risks, limitations, and future enhancements.